

Participant Information and Consent Form

Augmented Reality system for immersive training

Dear participant,

Thank you for taking the time to participate in this study. The following information explains the objective, procedure, voluntary nature of participation, and data protection measures of this study.

- **Study Objective**

The objective of this study is to examine the feasibility of an Augmented Reality system for immersive training on energy-saving behavior among building users.

- **Study Procedure**

First, you will complete a pre-experimental questionnaire. This questionnaire asks about your previous experience with immersive virtual environments and general demographic information.

Second, you will enter the AR environment in the iCARE lab (*Individual Comfort in Augmented Reality Environments*). After an adaptation phase, you will complete two attempts as part of an immersive training activity.

Third, after completing the training activity, you will fill in a post-experimental questionnaire about your experience with the system.

The total duration of the study is approximately 45 minutes.

- **Voluntary Participation and Right to Withdraw**

Participation in this study is completely voluntary.

You are free to decide whether you would like to participate in the study. If you decide not to participate, you will not experience any negative consequences.

You may stop participating in the study at any time without giving a reason. If you decide to withdraw during the experiment, you may simply inform the researcher that you would like to stop. You do not need to explain your decision.

- **Data Protection Statement**

All research data in this study will be collected fully anonymously. No directly identifying personal data, such as your name, email address, student ID, address, signature, or IP address, will be collected.

The collected data will be used exclusively for scientific research purposes and analyzed only in aggregated form, so that no conclusions can be drawn about individual participants.

The anonymous data will be stored securely and will only be accessible to authorized members of the research team.

The anonymous research data will be retained for 10 years and deleted afterwards.

Because the data are collected fully anonymously and no identifying information is stored, it will not be possible to identify or delete an individual dataset after completion of the experiment. The entire anonymous dataset will be deleted after the 10-year retention period.

- **Informed Consent**

Please confirm the following statements before continuing with the study:

- I have read and understood the study information.
- I understand that no directly identifying personal data will be collected.
- I have read and understood the information regarding data protection and agree to the handling of my data as described.
- I understand that the AR experience may cause symptoms of cybersickness, such as headache, eye strain, or general discomfort. I will immediately inform the experimenter if I experience any discomfort, even if the symptoms are only mild. In this case, the experiment will be paused or terminated. If necessary, first-aid assistance will be provided.

By participating in this study, you confirm that you have read and understood the information above and agree to the conditions described.

Date: _____

Participant's signature: _____

- **Contact Information**

If you have any questions or concerns regarding this study, please contact:

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For questions regarding data protection, you may also contact the Data Protection Officer of RWTH Aachen University:

Data Protection Officer:

dsb@rwth-aachen.de

Thank you for your time and your participation!

Kind regards,

Fan Ye

Qirui Huang

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